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PRECAMBRIAN PALAEOMAGNETISM IN AUSTRALIA I: BASIC DYKES AND VOLCANICS FROM THE YILGARN BLOCK

J.W. GIDDINGS

Research School of Earth Sciences, Australian National University, Canberra, A.C.T. (Australia)

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ABSTRACT

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The palaeomagnetism of 54 dykes sampled from the many swarms that intrude the Yilgarn Block of Western Australia shows that their directions of magnetization fall into five distinct groups.

Preliminary Rb—Sr age determinations made on samples used in the study however, indicate that for one, and possibly two of those groups, the dykes belong to two widely different ages suggesting that the palaeomagnetic field has been sampled at two widely separated points in time but when its direction was essentially the same. The seven groups of directions thus obtained imply that the dykes belong to at least six, possibly seven periods of intrusion. Relative-age information relating to four of those periods, determined from cross-cutting and baked contact studies, agrees with that obtained radiometrically.

The poles and ages of the groups are as follows:

Group YA dykes (c. 2500 or c. 1700 m.y.)	21.7S	133.7E	(A_{95} = 17.9°)
Group YB dykes (750—700 m.y.)	19.9S	282.0E	(A_{95} = 28.1°)
Group YC dykes ($\leq$1500 m.y., > Group YB)	79.7N	2.7E	(A_{95} = 13.0°)
Group YD dykes ($\geq$1700 m.y.)	23.5S	46.1E	(A_{95} = 9.9°)
Group YE dykes (c. 2500 m.y.)	28.3S	0.4E	(A_{95} = 31.0°)
Group YF dykes (c. 1700 m.y.)	24.7N	101.8E	(A_{95} = 14.0°)
Ravensthorpe dykes (2500 ± 100 m.y.)	38.3S	136.2E	(A_{95} = 25.5°)

Results from eight samples of the 1390 ± 140 m.y. Morawa Lavas of the Billeranga Hills yield a pole at 42.8N 22.4E (A_{95} = 14.7°).

INTRODUCTION

The possibility of correlating igneous rocks over wide areas using the palaeomagnetic technique has been demonstrated by palaeomagnetic studies in both the Canadian Shield (Fahrig et al., 1965; Larochelle, 1966) and

the Southern African Shield (McElhinny and Opdyke, 1964; Jones and McElhinny, 1966). The technique is based on the presumption that, for a particular structural element, rocks of the same age will have similar directions of magnetization whereas rocks of differing age will have different directions. It is particularly useful in establishing, in the absence of field evidence, whether or not the dyke swarms or other intrusives of a certain area belong to more than one age. The technique has been applied to the many dyke swarms that intrude the Yilgarn Block of Western Australia. The purpose was firstly to establish the age groupings of the dykes collected and secondly to determine the absolute age of each group from isotopic measurements on the samples belonging to each group. The parallel Rb—Sr study of those samples is being reported elsewhere (Compston and Crawford, pers. comm., 1975). The Rb—Sr age determinations may ultimately allow the use of the palaeomagnetic data to help define a Precambrian apparent

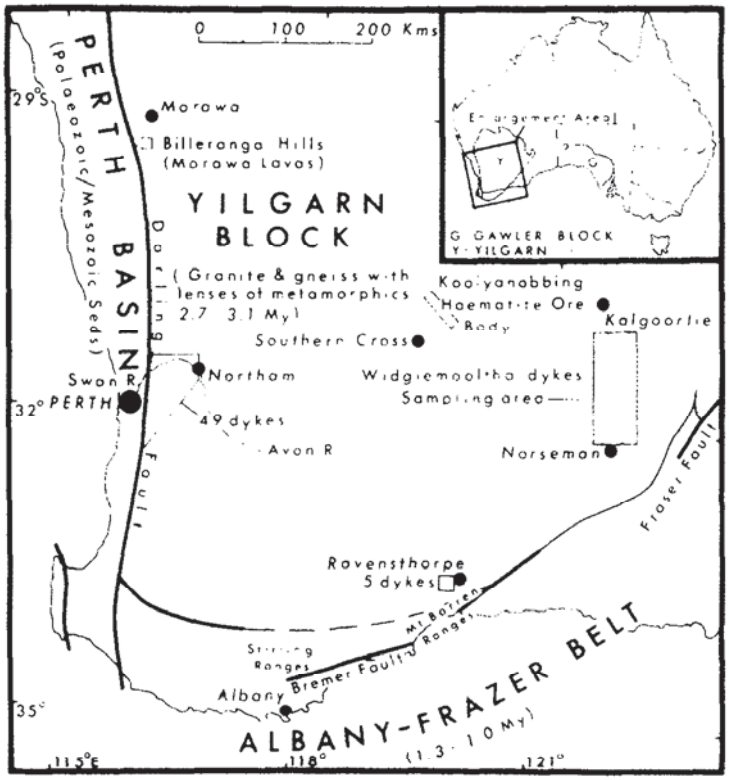


Fig. 1. The southern Yilgarn Block, showing the sampling areas of the dykes and volcanics, the location of the Widgiemooltha dyke suite (studied by Evans, 1968) and the haematite ore bodies of the Koolyanobbing Hills (studied by Porath and Chamalaun, 1968).

polar wander path for Australia. In this context, a result is also reported from the 1400 m.y. (1390 ± 140 m.y. — P.A. Arriens, unpublished information) Morawa Lavas of the Billeranga Hills (Arriens and Lalor, 1959; Compston and Arriens, 1968).

Prior to the present investigation, the only systematic study of Precambrian rocks in Australia was of the various haematite ore bodies which outcrop in South and Western Australia (Chamalaun and Porath, 1968; Porath and Chamalaun, 1968). Unfortunately the absolute ages of those ore bodies are not known. The ore bodies are of secondary origin (the products of subsequent enrichment of the Archaean and Proterozoic banded iron formations in which they are found) and are not amenable to any form of isotopic dating. Only two of the ten results obtained from the ore-bodies study refer to the Yilgarn Block, specifically to the ore bodies developed in the Archaean banded iron formations of the Koolyanobbing Hills (Fig. 1). Apart from those two results, the only other Precambrian result from this Block is that for the 2420 ± 30 m.y. Widgiemooltha dykes (Evans, 1968) which outcrop in the region between Kalgoorlie and Norseman (Fig. 1).

GEOLOGY

The Yilgarn Block of southwestern Australia (Fig. 1) is the largest of a number of ancient Precambrian crustal nuclei in Australia which collectively form the Australian Platform (Hills, 1965). It extends over an area of 900 km by 700 km and consists mainly of Archaean granite and gneiss in which older NNW-trending metasediments and metavolcanics are preserved (Prider, 1965; Compston and Arriens, 1968). Three periods of acid magmatism and metamorphism have been distinguished by Rb—Sr geochronology: at 3100—2900 m.y., 2700—2550 m.y. and 2300—2200 m.y. (Arriens, 1971). Most of the granites and many of the gneisses were emplaced between 2700 m.y. and 2600 m.y. The Darling Fault forms the western margin of the Yilgarn Block. It is a major structural lineament extending meridionally for 1000 km and takes the form of a prominent scarp which rises up to 300 m above the coastal plain to the west. The fault zone has been the locus of intermittent tectonic activity throughout the Phanerozoic (McWhae et al., 1958) and possibly the Precambrian (Wilson, 1958; Wilson et al., 1960), although it is currently seismically inactive (Doyle et al., 1968). To the south and southeast the Yilgarn Block is bounded by the Albany—Frazer Belt, a zone of rocks which were metamorphosed between 1300 and 1000 m.y. ago.

Many swarms of dykes intrude the various Precambrian formations of the Yilgarn Block. They are comparatively rare in the interior of the Block. Outcrop there is generally poor due to peneplanation of the land surface in the Late Tertiary (Jutson, 1934) and erosion under arid conditions in recent times. In the case of the Widgiemooltha dykes (Fig. 1) however,

extensive sampling is possible and results from those dykes have already been reported by Evans (1968). By contrast, dykes are ubiquitous along the western margin of the Block adjacent to the Darling Fault (Prider, 1965) and along its southern margin just north of the Albany—Frazer Belt (Sofoulis, 1958). In the present investigation therefore, attention was confined to the marginal dyke swarms, sampling of which was undertaken in the vicinity of Perth (western margin dykes) and near Ravensthorpe (southern margin dykes).

The basic dykes of the Perth region invade all rocks of the area and have generally been classified as either quartz-dolerites or epidiorites. The term 'epidiorite' has been used to refer to those quartz-dolerites which have suffered varying degrees of uralitization (Miles, 1938), an alteration which is considered to have resulted from deuteric processes rather than later regional metamorphism (Fletcher and Hobson, 1932; Cole and Gloe, 1940; Davis, 1942; Prider, 1944; Geary, 1952). The majority trend between NW and NE and have been injected along zones of weakness such as bedding planes in the metasediments and joints in the granite (Miles, 1938; Cole and Gloe, 1940) or parallel to shear zones in the country rock (Prider, 1948a). The intrusive network is particularly dense in the vicinity of the Darling Fault, where crustal extension due to intrusion has been estimated at 10% (Prider, 1948b), but falls off inland. There is some field evidence to indicate that the dykes belong to more than one episode of intrusion (Prider, 1943; Thomson, 1943; Stephenson, 1970), but the number of episodes is unknown. Indirect evidence from radiometric dating (Wilson et al., 1960; Compston and Pidgeon, 1962) suggested that the age of intrusive activity was Late Precambrian, between 750 m.y. and 500 m.y. (Prider, 1965; Compston and Arriens, 1968).

In the Ravensthorpe area, the Archaean basement consists of a remnant, arcuate structure of synclinally folded metasediments and metavolcanics preserved in a sea of gneiss (Sofoulis, 1958). A magmatic granite, emplaced within the core of the arcuate structure, is intruded by NNW-trending basic dykes which have been converted to amphibolites. They have been interpreted as end-stage differentiates of the consolidating magma, a pegmatite of which gives an age of about 2700 m.y. (Jeffery, 1956). To the south, a narrow belt of folded, stratified metasediments outcrops in the Mt. Barren Ranges. A similar series finds topographic expression in the Stirling Ranges about 275 km to the west, north of Albany (Fig. 1). Sofoulis (1958) has concluded that the Stirling Ranges metasediments are a westerly extension of those in the Mt. Barren Ranges. The Stirling Ranges metasediments have been radiometrically dated (Turek and Stephenson, 1966) and give a metamorphic age of 1150 m.y. and a minimum depositional age of 1340 m.y. The youngest known intrusives in the area are a swarm of E—ENE-trending quartz-dolerite dykes. They cut all rocks of the Archaean basement but do not penetrate the metasediments of the Mt. Barren Ranges. Accepting the conclusion of Sofoulis regarding the equivalence of the two

metasedimentary sequences, the age of those dykes would lie in the range 2700 m.y.—1340 m.y.

The Billeranga Hills (Fig. 1) are situated about 300 km north of Perth just inland of the Darling Fault. They form part of the outcrop of the Billerana Group (McWhae et al., 1958; Arriens and Lalor, 1959), a sedimentary and interbedded volcanic rock sequence which unconformably overlies the Archaean basement gneisses. The succession is gently tilted to the west and is intruded by dolerite dykes of at least two ages. The interbedded Morawa Lavas are of trachytic and andesitic composition and give (Compston and Arriens, 1968) a Rb—Sr isochron age of about 1400 m.y. (1390 ± 140 m.y. — P.A. Arriens, unpublished information).

SAMPLING DETAILS

Forty-nine dykes from the Perth region and five dykes from the Ravensthorpe district (Fig. 1) were sampled and ten samples were taken from a 30-m section of the Morawa Lavas in the Billeranga Hills. The dykes varied in width from 3 m to 30 m.

In the Perth region dykes were sampled in two areas. Away from the Darling Fault zone, nineteen dykes were sampled from fresh exposures in a series of rail cuttings along the Avon River Valley northeast of Perth (Fig. 1), as far inland as Northam. The other thirty dykes were sampled in the vicinity of the Darling Fault zone, east of Perth. The dykes were freshly exposed and all but two were sampled from quarries. In one quarry samples were taken from a pair of dykes, one of which cuts the other. In another quarry a pair were sampled which were in contact but where any cross-cutting relationship that might have been present was obscured by recent quarrying operations. Six samples were collected of baked contact material adjacent to three dykes.

The five dykes sampled in the Ravensthorpe district belong to the younger suite of E—ENE-trending dykes and were collected, by field drill, from fresh exposures in the bed of Phillips River, about 20 km west of Ravensthorpe. Attempts to sample the baked contact zones of the country rock gneisses were unsuccessful.

Except for the Ravensthorpe dykes, block sampling was employed. Wherever possible, 3—5 samples were generally taken from each dyke distributed across the width. A sun compass was used for orientation purposes, checked with the aid of a magnetic compass. In the laboratory, three cylindrical specimens (28 mm × 28 mm) were generally cut from each sample or core. The 205 samples yielded 610 specimens.

PALAEOMAGNETIC MEASUREMENTS

The direction and intensity of magnetization of specimens were measured using a PAR model SM1 spinner magnetometer. Directions have been ana-

TABLE I

Dyke mean directions of remanence before and after magnetic cleaning

Locality	Dyke No.	Before cleaning					After cleaning				
		N	R	$D(^\circ)$	$I(^\circ)$	Int (mA.m^{-1})	$n(N)$	R'	$D(^\circ)$	$I(^\circ)$	group
Ravensthorpe dykes											
Phillips River	1	2	1.70	–	–	563	16(5)	15.69	91.3	77.6	RD
	2	2	1.78	–	–	1356	6(2)	5.88	129.1	73.2	RD
	3	2	1.99	36.3	33.2	1020	6(2)	5.98	10.0	81.4	RD
	4	2	1.96	203.1	39.7	994	6(2)	5.98	188.0	80.7	RD
	5	2	1.39	–	–	843	3(1)*	2.94	285.9	−51.3	–
Perth region dykes											
(1) Avon Valley											
Rail cutting no. 3	8	5	3.20	–	–	0.5	7(3)	6.05	347.3	49.6	YF
	9	3	2.91	359.4	26.7	18.3	unstable.				
Rail cutting no. 4	10	5	4.00	174.7	−86.3	38.0	10(4)	9.12	230.2	−82.5	YA
	11	3	2.93	20.6	−49.5	11.4	9(3)	8.90	338.5	−38.2	YC
Rail cutting no. 5	12	3	2.48	–	–	120	6(2)	5.44	10.3	48.6	YF
	13	3	2.72	345.8	−2.0	3.8	7(3)	6.73	333.3	49.0	YF
Rail cutting no. 6	14	5	3.01	–	–	120	14(5)	13.62	5.6	45.5	YF
Rail cutting no. 7	15	5	4.70	11.0	34.2	671	15(5)	14.90	16.9	51.3	YB
Rail cutting no. 8	16	3	2.77	156.0	−63.8	346	unstable.				
	17	4	3.94	354.8	−56.7	311	9(4)	8.32	357.4	43.0	YF
Rail cutting no. 9	18	2	1.99	191.1	−26.3	66.5	6(2)	5.94	223.6	18.4	YE
	19	3	2.98	167.8	−46.2	40.4	7(3)	6.96	170.1	−40.9	YF
Rail cutting no. 10	20	2	1.92	169.7	−71.2	52.9	3(2)	2.92	167.0	−44.7	YF
	21	3	2.98	160.8	−62.6	93.6	7(3)	6.65	168.5	−39.3	YB
	22	3	2.67	248.0	−62.9	249	6(2)	5.93	224.6	−11.5	YE
Rail cutting no. 11	23	3	2.84	20.8	−15.9	60.0	9(3)	8.78	17.3	−45.3	YC
	24	3	2.13	–	–	42.3	6(2)	5.73	113.6	−53.1	YF
Rail cutting no. 12	25	2	1.97	350.9	27.8	20.8	3(1)*	2.83	353.4	45.9	YF
Quarry	26	3	2.88	139.9	73.9	62.3	9(3)	8.92	135.3	56.5	YB
(2) Darling Scarp											
Rail cutting no. 1	6	3	2.78	110.6	2.4	0.4	7(3)	6.28	255.0	52.7	YD
Rail cutting no. 2	7	3	1.45	–	–	88.3	6(2)	5.79	246.7	39.4	YD
Canning Dam Quarry	27	3	2.81	229.1	56.7	104	5(2)	4.93	188.7	56.8	YF

TABLE I (continued)

Locality	Dyke No.	Before cleaning					After cleaning				
		N	R	$D(°)$	$I(°)$	Int ($mA.m^{-1}$)	$n(N)$	R'	$D(°)$	$I(°)$	group
Darlington Quarry	28	3	2.21	—	—	99.4	9(3)	8.83	91.1	78.9	YA
	29	3	2.39	—	—	112	8(3)	7.65	37.0	75.4	YA
	30	2	1.97	193.9	−66.8	45.6	6(2)	5.87	122.1	−51.4	YF
Boya Quarry	31	5	2.43	—	—	54.1	11(4)	10.32	328.5	64.2	YB
	32	5	4.91	36.9	−72.6	2.9	15(5)	14.68	345.1	−74.2	YC
Greenmount Quarry	33	6	5.63	31.7	70.7	192	18(6)	17.90	50.5	82.9	YA
	34	2	1.81	—	—	8.3	9(3)	8.50	291.4	80.9	YA
Serpentine Dam	35	5	4.74	246.4	−75.8	603	12(4)	11.42	263.8	−62.9	YA
	36	3	2.81	247.7	43.6	430	9(3)	7.71	250.4	30.5	YE
Stathams Quarry	37	3	1.79	—	—	34.0	7(4)	6.75	272.2	64.2	YD
	38	2	1.95	35.4	−54.7	1.1	6(2)	5.81	23.7	−62.2	YC
Readymix Quarry	39	5	4.23	304.1	−33.4	14.7	10(4)	9.27	281.6	47.2	YD
	40	3	2.28	—	—	85.5	6(2)	5.84	243.8	30.7	YD
	41	3	2.53	—	—	7.7	partially remagnetized.				
	42	3	1.81	—	—	71.8	6(2)	5.95	273.2	43.4	YD
	43	3	1.66	—		0.8	8(3)	7.24	266.5	63.9	YD
Swan Quarry	44	5	3.71	84.9	−18.8	25.5	12(5)	11.35	173.3	63.6	YC
	45	2	1.90	—		0.8	6(2)	5.57	320.2	−50.8	YC
	46	3	2.60	—	—	27.2	4(2)	3.84	352.0	−35.8	YC
	47	2	1.75	—	—	2.0	6(2)	5.60	254.6	42.4	YD
	48	2	1.86	—	—	0.9	partially remagnetized.				
	49	4	3.54	351.7	−30.1	7.6	11(4)	9.92	2.3	−43.8	YC
	50	5	4.94	250.8	25.0	5.3	18(6)	17.87	257.8	28.1	YD
Pioneer Quarry	51	3	2.92	335.0	18.9	27.9	9(3)	8.76	324.5	35.2	YC
	52	5	4.53	357.8	−51.5	97.9	11(5)	9.73	4.5	−59.9	YC
	53	3	2.86	45.1	−38.3	1.5	4(2)	3.97	60.5	−36.2	YD
	54	2	1.85	—	—	42.0	8(3)	7.60	318.3	−53.0	YC

Notes: (1) N, n = number of samples, specimens per dyke and R, R' = length of resultant of N, n unit vectors. $D(°)$ and $I(°)$ = declination and inclination of mean direction of magnetization and $Int.$ = intensity of magnetization.

(2) Directional data are omitted for those dykes whose magnetization is random using the test for randomness of directions (Watson, 1956; Vincenz and Bruckshaw, 1960).

* Dyke-mean is based on specimens from only 1 sample. The direction has been excluded from paleomagnetic pole calculations.

lyzed using the statistics developed by Fisher (1953). Alternating field (AF) demagnetization was performed with equipment described by Evans (1969), based upon the principles given in McElhinny (1966). Stepwise demagnetization of pilot specimens chosen from the dykes and lavas, was carried out at seventeen field values (where necessary) ranging from 5 mT to 210 mT, the maximum available.

The initial remanence (NRM) directions of the dykes were scattered (Table I). However, this is a feature that has often been observed in previous studies of Precambrian dykes and sills (McElhinny and Opdyke, 1964; Larochelle, 1966). Repeat measurements of the NRM of one specimen from each sample after an elapsed period of three months showed that two-thirds of the collection had acquired temporary components of magnetization (Creer, 1957). Angular changes of 100° and intensity changes of 200% were observed in extreme cases.

The stability of remanence was tested by subjecting at least two pilot specimens per dyke to AF demagnetization. The stability characteristics of the collection have been expressed in terms of the Stability Index (PSI) defined by Giddings and McElhinny (in press). The PSI values determined for the 112 pilot specimens are shown in histogram form in Fig. 2. Although the indices obtained span the complete spectrum, there is a marked concentration at low values. About two-thirds of the pilot specimens showed good end-point behaviour with higher values of the Briden optimum cleaning index (Briden, 1972). The remainder were either partially stable or unstable. The

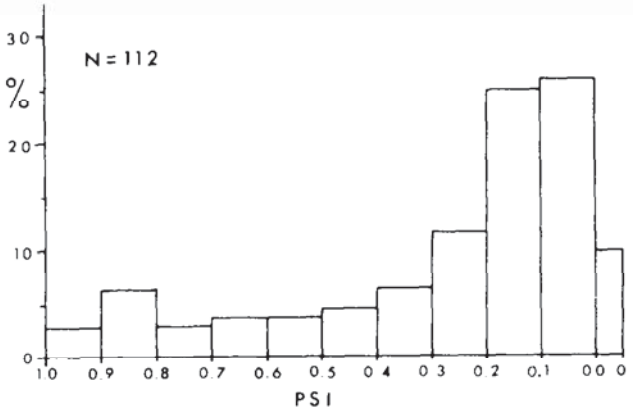


Fig. 2. The histogram summarizes the stability characteristics of the collection of samples from dykes, expressed in terms of the stability index (PSI) defined by Giddings and McElhinny (in press). The distribution of PSI values shows that the remanence of the dykes spans the complete stability spectrum, from extremely stable remanence (PSI > 0.80) to partially stable and unstable remanence (PSI < 0.11). The peak at low values, however, indicates that the general level of stability is low.

stability characteristics of the collection are in marked contrast to those of the South Australian dykes (Giddings and Embleton, 1976) whose indices peak at high values in addition to low values. Cleaning fields generally lay between 5 mT and 32.5 mT. An interesting feature was the wide variation in PSI values within some of the dykes. For about half the dykes one pilot specimen gave PSI values >0.11 (stable to extremely stable remanence) whereas another from the same dyke proved to be partially stable or unstable. This necessitated examination of the cleaned data at the specimen rather than the sample level. For statistical purposes therefore, each specimen was given unit weight in calculation of dyke-mean directions of cleaned remanence (Table I).

Forty-eight of the dykes responded satisfactorily to magnetic cleaning. The mean directions of magnetization from dykes of the Perth region fall into five distinct groups; the Ravensthorpe dykes belong to a single group (Fig. 3). The different groups imply that the dykes belong to different ages of intrusion. Samples from each of the groups were submitted for Rb—Sr age determination and the geochronological study confirmed that the groups of dykes were widely different in age (Compston and Crawford, pers. comm., 1975) (Fig. 3 and Table III). In addition, the following two points emerged:

(1) The Perth region YA group of dykes and the Ravensthorpe dykes could be similar in age as suggested by their similar directions of magnetization (Fig. 3). Thus, one interpretation of the geochronological data from the YA group of dykes is that they are around 2500 m.y. old, compared with the 2500 ± 100 m.y. age for the Ravensthorpe dykes. However, an alternative interpretation of that data suggests that those dykes are around 1700 m.y. old. Further work is being undertaken to resolve the ambiguity. Until more evidence becomes available favouring either one or other of the ages, the distinction between the two groups of directions is retained.

(2) The similar directions of magnetization of groups YB and YF (Fig. 3) suggested that those dykes belonged to the same swarm. Dating revealed however, that the dykes belong to two swarms of widely different ages. The dykes have therefore been separated into the two groups according to the additional, geochronological information.

At least six, possibly seven, groups of dykes of differing age are thus represented by the collection. The trends of the dykes within each group are shown in Fig. 3. No group of Perth region dykes is represented by a unique trend and all but one of those groups possesses both normal and reversed polarities of magnetization.

Sample-mean directions of NRM of the Morawa Lavas are rather scattered (Table II and Fig. 4). Stability of remanence was tested by subjecting six specimens to stepwise AF demagnetization and two specimens to partial thermal demagnetization up to a maximum temperature of 640°C. Pilot specimens subjected to AF demagnetization either remained stable up to the maximum field of 210 mT, or stabilized after an initial angular change which was accompanied by a large intensity drop. Grouping of

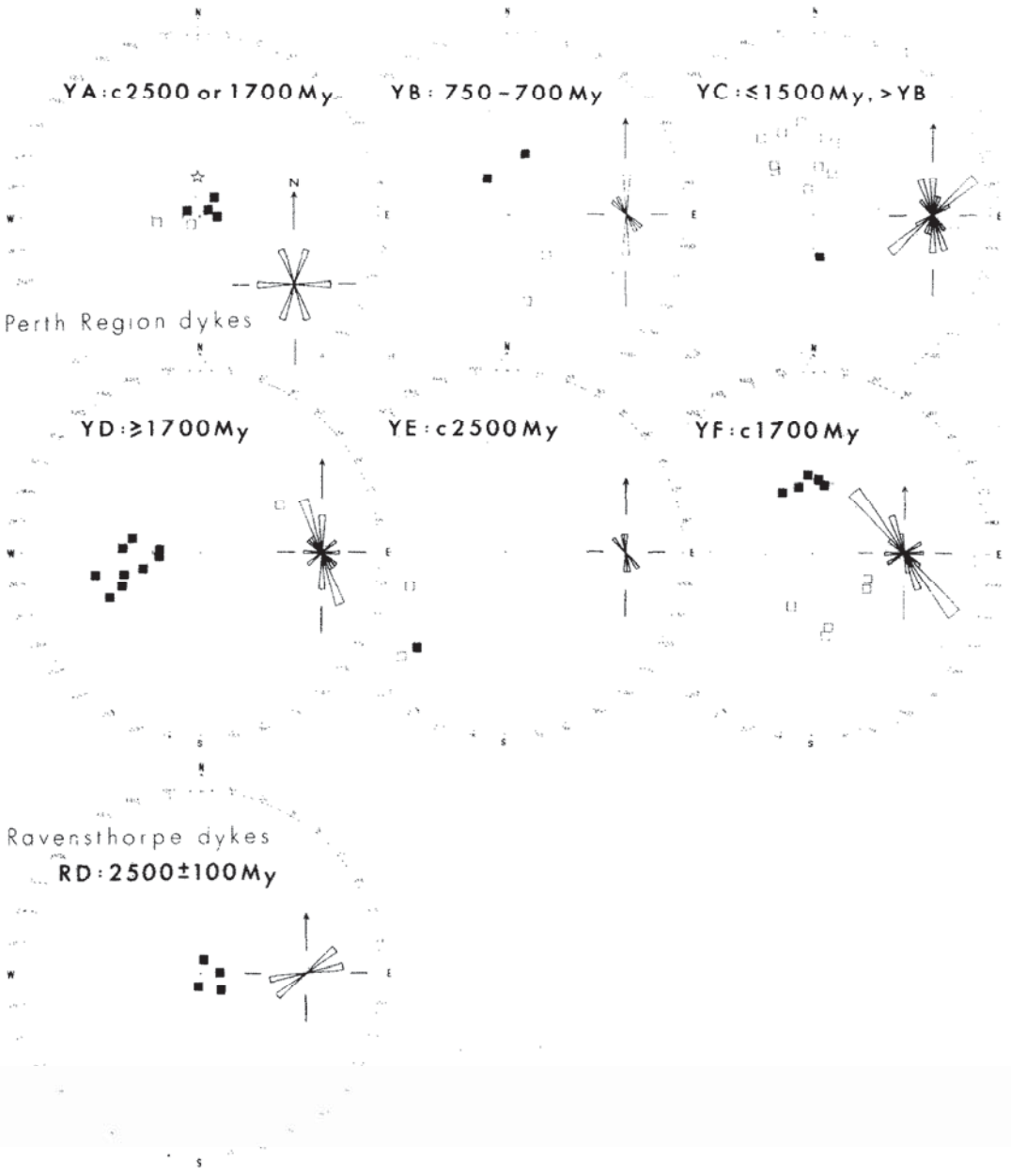


Fig. 3. Dyke-mean directions of magnetization after magnetic cleaning, grouped according to directional similarity and age information from the Rb—Sr dating study (see text). The compass rose diagrams depict the trends of the dykes within each group. The star represents the direction of the earth's present magnetic field in the area. Open (closed) symbols denote upward (downward) pointing directions. They are plotted on Wulff equal-angle projections.

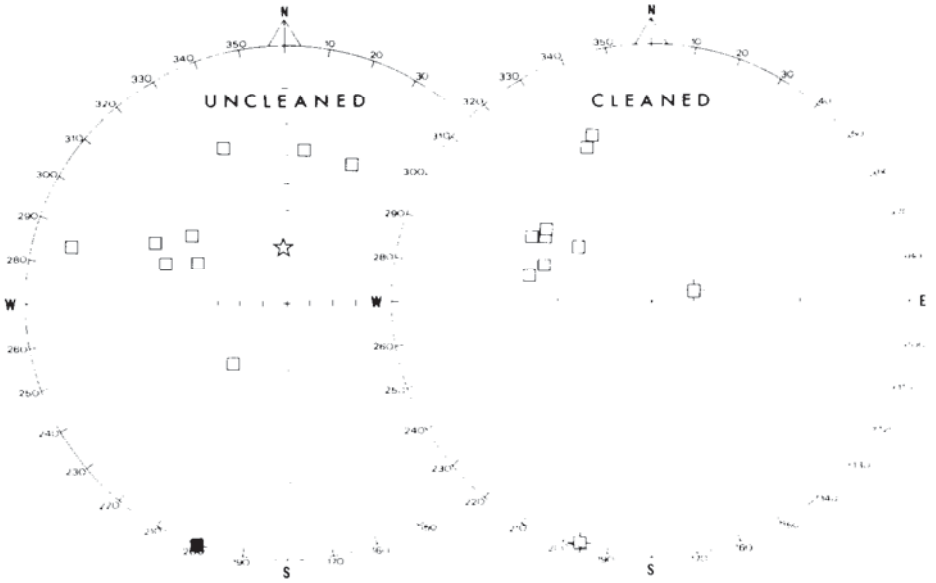


Fig. 4. Morawa Lavas: Sample-mean directions of magnetization before and after magnetic cleaning. Crossed symbols denote sample-mean directions omitted from calculation of the formation-mean direction and pole position. The direction of the earth's present magnetic field in the area is represented by a star. Symbol plotting convention and projections as for Fig. 3.

TABLE II

Morawa Lavas: Sample-mean directions of magnetization before and after magnetic cleaning

Sample	Before cleaning					After cleaning			
	N[1]	R	D(°)	I(°)	Int(mA. m^{-1})	N	R	D(°)	I(°)
BH 1	3	2.999	306.1	−41.7	46.9	3	2.999	304.9	−38.3
BH 2	3	2.997	288.9	−38.7	20.8	3	2.997	300.6	−39.0
BH 3	2	1.999	294.9	−48.8	14.6	2	1.998	288.7	−42.5
BH 4	3	2.999	295.2	−32.0	18.2	3	2.999	299.1	−35.2
BH 5	3	2.988	220.6	−54.6	5.4	3	2.903	340.4	−21.4
BH 6	3	2.988	338.5	−24.3	83924	3	2.990	338.0	−23.5
BH 7	3	2.999	285.2	−9.7	3527	3	2.991	282.2	−39.1
BH 8	3	2.999	25.4	−28.2	277	3	2.999	307.4	−51.2
BH 9[2]	3	2.991	6.6	−27.7	15.7	3	2.822	78.0	−71.5
BH 10[2]	3	2.994	199.6	0.1	830	3	2.989	195.9	−1.2

Notes: (1) N = number of specimens. Other symbols explained in Table I.
(2) Sample-mean direction not used in the calculation of the formation-mean direction or pole position (see text).

pilot directions improved after AF demagnetization due to removal of soft, secondary, viscous magnetization. All remaining specimens were cleaned in a field of 20 mT. The sample-mean directions of cleaned remanence (Table II) are shown in Fig. 4. While cleaning reduced directional dispersion between eight of the samples, the magnetization of the remaining two samples failed to respond and their directions remained oblique to the main population. They may represent real oblique (intermediate?) directions or, more likely, samples of material not properly in situ. Those two samples have been excluded from calculation of the pole position. Directions from the two thermally demagnetized pilots remained stable up to 640°C and agree with the main population directions derived from AF studies. The thermal results additionally indicate that in two samples at least, haematite is the source of stable remanence. Both magnetite and haematite are present in the lavas (Arriens and Lalor, 1959). The haematite is considered however, to date from the time of eruption of the lavas and to have formed during their sub-aqueous extrusion.

BAKED CONTACT STUDIES

Baked contact studies proved useful as a field test for establishing the time-stability of remanence (McElhinny, 1973) and for establishing on palaeomagnetic evidence alone, the relative ages of some of the groups of dykes.

The magnetizations of the fifteen specimens from the six samples of baked granite—gneiss country rock were at the lower limit of measurement ($0.1—0.3\ mA.m^{-1} = 1—3 \cdot 10^{-7}$ G) of the magnetometer and too inhomogeneously magnetized to produce meaningful results. Further work on them was not pursued. In many quarries and rail cuttings dykes belonging to more than one group are present (Table I). Three instances of a later dyke partially baking an earlier dyke were found and are shown in Figs. 5 and 6. Fig. 5a illustrates the situation found in one quarry where four dykes are exposed in one wall as shown. Three of the dykes belong to group YD and one to group YC. The relationship of the YC group dyke to the YD group dyke was obscured in the field because of recent quarrying operations. However, as the illustration shows, a sample from the extreme margin of the YC group dyke at its interface with a YD group dyke records a YC group direction. This demonstrates that the YC group dyke is younger than the YD group dyke. As a result of intrusion of the younger YC group dyke, another adjacent YD group dyke has been partially thermally remagnetized and records both YC group and YD group directions across its width. In another quarry (Fig. 5b) a YC group dyke is cut by a YB group dyke. Samples from the margins of the YC group dyke, where in contact with the YB group dyke, record YB group directions. This clearly demonstrates that those margins were thermally remagnetized when the YB group dyke was intruded and furthermore that the YC group of direc-

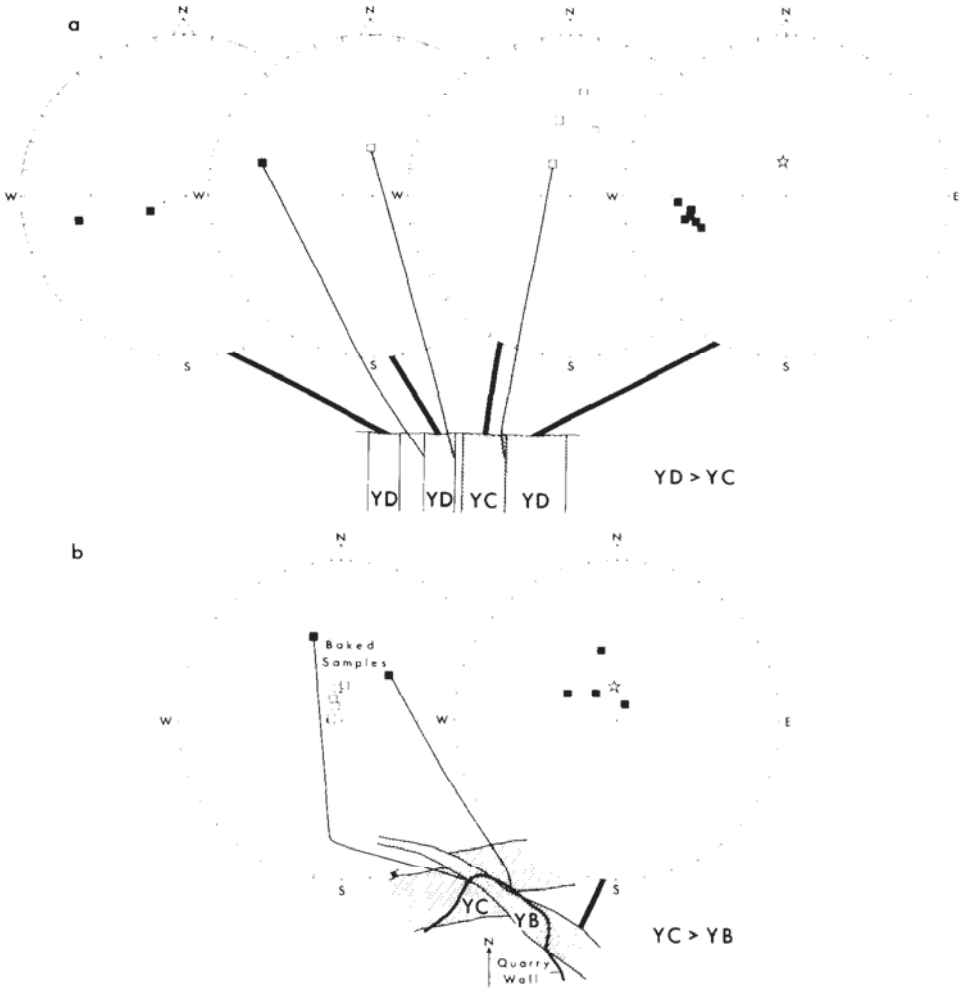


Fig. 5. Baked contact relationships observed in dykes from two quarries in the Perth region. In (a) the group YC dyke has partially remagnetized an adjacent group YD dyke. In (b) the group YC dyke has been thermally remagnetized by the group YB dyke along its zones of contact with the group YB dyke. The relative ages of the three groups of dykes, determined from these results, are shown.

tions, despite being close to the present earth's field direction in the area, were acquired prior to intrusion of the YB group of dykes. Finally, the situation depicted in Fig. 6 enables the age relationships to be inferred. Traversing the width of the dyke (right to left in Fig. 6), directions typical of the YA group pass, via intermediate directions, to those typical of the YC group. The YA pilot response suggests that the useful limit of the coercivity spectrum is about 40 mT. The YC pilot specimen direction rotates

TABLE III

Summary of group-mean directions of magnetization, ages and pole positions from the dykes and lavas

	Group	Age (m.y.)	N[1]	R	D(°)	I(°)	α_{95}(°)	Pole position		A_{95}(°)
								lat (°)	long (°)	
Ravensthorpe Dykes	RD	2500 ± 100	4	3.94	114.3	82.9	13.2	38.3S	136.2E	25.5
Perth Region Dykes	YA	c.2500 or c.1700[2]	6	5.89	61.1	80.6	10.0	21.7S	133.7E	17.9
	YB	750—700	4	3.84	164.4	−55.0	21.7	19.9S	282.0E	28.1
	YC	≤1500, > group YB[3]	11	10.47	347.7	−53.0	10.8	79.7N	2.7E	13.0
	YD	≥1700[4]	10	9.66	257.6	45.5	9.5	23.5S	46.1E	9.9
	YE	c.2500	3	2.76	232.1	−8.1	46.1	28.3S	0.4E	31.0
	YF	c.1700	10	9.59	344.9	50.8	10.5	24.7N	101.8E	14.0
Morawa Lavas	ML	1390 ± 140	8	7.58	309.0	−38.1	14.0	42.8N	22.4E	14.7

Notes: (1) N = number of dykes or in the case of the lavas, number of samples. α_{95}, A_{95} = semi-angle of the cone of confidence about the mean direction of magnetization, mean pole at the 95% probability level (Fisher, 1953).

(2) The Rb—Sr isotopic data are consistent with either of these approximate ages, depending on how the data are interpreted.

(3) The maximum age of the dykes is 1500 m.y. They may be younger. The cutting of a group YC dyke by a group YB dyke (Fig. 5b) yields a lower limit on their age.

(4) This is the minimum age. The dykes may be older.

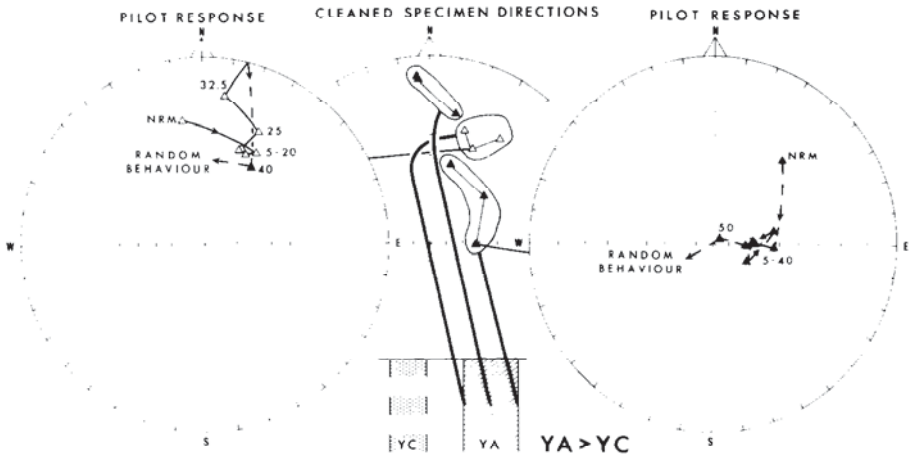


Fig. 6. Pilot specimen behaviour and directions of cleaned magnetic remanence observed in specimens from a partially remagnetized dyke from a quarry in the Perth region. The results permit the inference that group YA dykes are older than group YC dykes, (see text for explanation). The numbers in the pilot response diagrams refer to the demagnetizing fields (in mT) used in stepwise alternating field demagnetization.

from its stable direction in low fields and approaches the direction of the YA pilot after demagnetization in a field of 40 mT. However, stabilization of its direction in better agreement with the YA pilot direction is not revealed at higher fields since they exceed the maximum coercivity. The behaviour of this pilot is taken to indicate that this dyke was originally a YA group dyke which has subsequently suffered partial reheating from an adjacent dyke inferred to belong to the YC group (not all dykes exposed in the quarries were sampled). The component present in the YC pilot and stable in lower fields is considered to be a viscous PTRM (Chamalaun, 1964) acquired during reheating. The ratio of that component to the original component varies across the width of the dyke and accounts for the streaking observed between the YC group and YA group directions.

From the baked contact studies, it is evident the following age relationships hold: YD > YC > YB and YA > YC. These relationships are corroborated by the Rb—Sr dating studies (Fig. 3 and Table III).

DISCUSSION AND POLE POSITIONS

The stable components of magnetization isolated from the dykes are regarded as primary for the following reasons:

(1) Baked contact studies.

(2) Consistent directions of magnetization from geographically widely separated collecting localities.

(3) Presence of directions belonging to more than one group at single

localities, precluding widespread remagnetization.

(4) Laboratory stability tests.

It follows that the reversals of magnetization observed in the different groups of directions are probably records of reversals of the ambient geomagnetic field during intrusion of the dykes. Tectonic movements which potentially could alter the primary attitudes of the dykes were probably restricted to epeirogenic tilting, warping and relative uplift associated with activity along the Darling Fault. The resulting shallow-angled differences in attitudes are of course difficult to detect and only by making geographically extensive collections within individual cratons can the effects of relative movement be eliminated. The stable remanence revealed in eight samples of the Morawa Lavas is considered primary on the basis of the laboratory stability tests and the fact that its pole (Table III) is different to poles so far determined for younger rocks (McElhinny, 1973).

Group-mean directions of magnetization, ages and pole positions for the various groups of dykes are given in Table III. The palaeomagnetic technique of separating igneous activity into discrete events of differing age was particularly useful in those cases where directions of remanence were different. However, the investigation has highlighted the point that when the timescale of such activity is very long, as in the present case where it covers approximately 1800 m.y., caution is required in arguing the converse, that similar ages can be assumed on the basis of similar directions of remanence. This is not surprising in view of the fact that over such a long time span, the average rate of apparent polar wander of $0.2°$—$0.3°$ m.y.$^{-1}$ (McElhinny, 1973) would enable the path of the pole to cross itself at least once, if not several times. That situation has arisen, for example, in the Precambrian apparent polar wander path for Africa (Piper et al., 1973).

Although six episodes of dyke intrusion have been distinguished in the Perth region, extending in age into the Archaean, the dating work of Compston and Pidgeon (1962) and Compston and Arriens (1968) suggests the presence there of further, very young, Precambrian swarms which are not represented in the present collection. A complex intrusive history is thus indicated for the Perth region and, by implication, the whole western margin of the Yilgarn Block. The similarity in dyke trends between the various swarms and their sub-parallelism to the Darling Fault suggests that the intrusive episodes may be related to periods of tectonic activity along that fault, believed to be an Early Precambrian feature (Wilson, 1958; Wilson et al., 1960).

The interpretation of the results represented in terms of Precambrian apparent polar wander is only possible when data from the Gawler Block (Giddings and Embleton, 1976), the Mount Isa Block (Duff and Embleton, in press) and the Kimberley Block (McElhinny and Evans, in press) are considered together. That analysis and the use of the data to determine the age of formation of the haematite ore bodies of the Koolyanobbing Hills are in preparation.

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